

The Fibonacci–Mellin Transform: Vandermonde Non-Isometry, Polar Readout, and Coherent Rank-One Reduction

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Abstract

For signals organised multiplicatively rather than additively — log-periodic, self-affine, or discretely scale-invariant — the natural tool is a Mellin transform on a multiplicative lattice. We introduce the Fibonacci–Mellin Transform (FMT), a finite Vandermonde frame on a golden-ratio scale lattice, and establish its algebraic structure. *Exact* results: the raw output splits into a two-sheet LOW/HIGH decomposition related entry-wise by an Artin phase twist through the identity $1 + \varphi^{-1} = \varphi$; the frame is non-tight, and its polar factor $U_\varphi = WV^* \in U(10)$ is a canonical unitary readout; and the five-register two-sheet space $\mathbb{C}^{10} = V_5 \otimes \mathbb{C}^2$ admits a unique D_5 -coherent rank-one line $|\Omega\rangle$, whose composition with U_φ defines a canonical coisometry $\mathcal{R}_\chi = (\langle\Omega| \otimes I_2)U_\varphi : \mathbb{C}^{10} \rightarrow \mathbb{C}^2$ with $\mathcal{R}_\chi \mathcal{R}_\chi^* = I_2$, $\text{rank}(\mathcal{R}_\chi^* \mathcal{R}_\chi) = 2$, and $\dim \ker \mathcal{R}_\chi = 8$, so the coherent observable quotient is canonically two-dimensional. *Numerically certified for* $N = 10$, in multiprecision arithmetic stable from 40 to 120 digits: a real diagonal reweighting cannot unitarise the frame; the condition number $\kappa = 4.70 \times 10^{18}$, the effective numerical rank, and tested phase non-commensurability. The frame is ill-conditioned enough that the specific polar factor is not determined by double precision, while the coisometry identities hold at any precision. From the two sheets we build gain-invariant scale-coherence diagnostics that separate the reported synthetic golden-structured signals from the reported harmonic controls. The FMT is a fixed, parameter-free scale-transfer frame; the construction assigns no physical meaning to the two retained components.

Keywords. Fibonacci–Mellin transform; scale-memory analysis; two-sheet decomposition; non-unitary frame; polar unitarization; scale-coherence diagnostic; gain-

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invariant readout.

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1 Introduction

1.1 Scale memory vs frequency memory

The DFT on $\mathbb{Z}_N$ answers the question: *which harmonic frequency is present?* It operates on a closed additive cycle $k \mapsto k + N \equiv k$ and resolves scalar $U(1)$ characters $e^{-2\pi i j k / N}$. Each bin $X_j \in \mathbb{C}$ carries a single phase and a single energy value.

A different question arises when a signal's structure is organised multiplicatively rather than additively: *where is the signal inside a φ -autosimilar scale hierarchy, and how did it advance from one scale level to the next?* The golden fine lattice $x_k = \varphi^{k/5}$ closes quasi-periodically:

$$k \mapsto k + 10 \implies x_{k+10} = \varphi^2 x_k \neq x_k. \quad (1)$$

After one full traversal of the 10-node lattice, the scale advances by $\varphi^2 \approx 2.618$ rather than returning to the start. This advance is the *scale memory* that the FMT preserves and converts into measurable phase-coherence features.

1.2 Signal classes where the FMT is relevant

Table 1 classifies signal classes by their suitability for FMT analysis. The FMT is useful when the signal's structure is multiplicative, log-periodic, self-affine, or discretely scale-invariant; it offers no advantage over the DFT for purely harmonic signals.

1.3 Motivating experiments

Dumitrescu et al. [2] realized a dynamical symmetry-protected topological phase (ED-SPT) by driving 10 ytterbium qubits with a Fibonacci pulse sequence. Xu et al. [3] demonstrated non-abelian braiding of Fibonacci anyons on a superconducting processor. These experiments motivate two distinct candidate domains: Fibonacci-timed drives in EDSPT systems, and Fibonacci-category braid structures. The present paper does not claim a direct validation on either dataset; this is left as open problem P3.

1.4 Main results

- (i) *LOW/HIGH two-sheet decomposition* (Theorem 3.1, §3): the raw FMT has $y_j = \text{LOW}_j$ and $y_{j+5} = \text{HIGH}_j$, where the HIGH row is the LOW row pointwise Artin-twisted before projection: $(\mathcal{F}_\varphi)_{j+5,k} = A_j(k)(\mathcal{F}_\varphi)_{j,k}$. GP(5,2) is used as an

Table 1: Domain applicability of FMT scale-memory analysis. *High*: FMT directly resolves the scale structure. *Medium*: FMT provides complementary scale-coherence features. *Low*: DFT is sufficient and optimal.

Signal class	Priority	Structural reason
Log-periodic signals	High	Oscillations at scales φ^m ; Mellin-native
Discrete scale invariance	High	$f(\lambda x) = \lambda^h f(x)$ for discrete λ near φ
Quasi-periodic drives	High	Fibonacci pulse sequences; φ -rational frequency support
OMA/SHM modal families	Candidate	Useful if diagnostic target is log-scale coherence loss; validation against MAC and SSI required
SAR/DEM fractal texture	Candidate	Plausible for self-affine textures; no benchmark in this paper
$1/f^\alpha$ noise	Medium	Scale coherence complements spectral slope estimate
Multifractal turbulence	Medium	FMT as compact scale-ordering feature
NDVI / vegetation texture	Medium	Hierarchical canopy structure; log-scale patterns
Pure harmonic signals	Low	DFT is optimal; FMT offers no sparsity advantage
Standard OFDM / QAM	Low	Additive periodic closure; DFT is the native transform

indexing graph for the two sheets. $\text{CT}(2, 2)$ is optional post-processing, not part of $\mathcal{F}_\varphi$.

- (ii) *$U(2)/SU(2)$ doublet structure* (§4): the doublet $\Psi_j = (\text{LOW}_j, \text{HIGH}_j)^T$ carries a $U(1) \times SU(2)$ -type representation; Stokes-like observables (S_x, S_y, S_z) provide scale-internal feature vectors for each frequency index j .
- (iii) *φ -memory cell* (Proposition 6.1, §6): $G_j(k) \in U(2)$ is index-dependent. No rational commensurability of $\delta_{j,k}$ with π is detected numerically on the tested lattice phases; the observed phase orbits are consistent with quasi-periodicity ($G_j^n \neq I$ for all tested n).
- (iv) *Diagonal unitarization obstruction* (Proposition 8.1, §8): no real diagonal weights unitarize $\mathcal{F}_\varphi$, numerically certified for $N = 10$.
- (v) *Polar unitary readout* (Theorem 9.2, §9): $U_\varphi = WV^* \in U(10)$, embeds into

$U(16)$, admits canonical POVM and Halmos block-encoding.

- (vi) *Scale-coherence algorithm* (Algorithm 1, §7): a complete seven-step pipeline from signal to feature vector.
- (vii) *Coherent rank-one reduction* (Theorem 10.2, §10): on $\mathbb{C}^{10} = V_5 \otimes \mathbb{C}^2$ the dihedral register action selects a unique invariant line $|\Omega\rangle$; composing it with the polar factor gives a canonical coisometry $\mathcal{R}_\chi : \mathbb{C}^{10} \rightarrow \mathbb{C}^2$ with $\mathcal{R}_\chi \mathcal{R}_\chi^* = I_2$ and $\dim \ker \mathcal{R}_\chi = 8$, so the coherent quotient $\mathbb{C}^{10}/\ker \mathcal{R}_\chi$ is two-dimensional.

What is not claimed. The FMT does not outperform the DFT in spectral sparsity: $G_{\text{FMT}} < G_{\text{DFT}}$ (Gini) in all benchmarks, because the 4.70° Vandermonde arc makes FMT basis functions nearly parallel. $R_{\text{FMT}}^2 \approx 1$ for harmonic signals too; the operative discriminant is the contrast ΔTV and ΔR^2 , not R_{FMT}^2 alone.

1.5 Logical position

The upstream projection–SVD construction [1] is the primary source for the quadratic Pincherle threshold α_Π , the rank $r = 5$, the logarithmic block V_5 , the metric K_Π , and the rational transfer representative H_Π .

The present paper does not rederive those objects. It constructs the finite Mellin/Vandermonde frame on the golden lattice, proves the LOW/HIGH two-sheet decomposition, constructs the polar unitary readout, and derives the coherent coisometry $\mathcal{R}_\chi : \mathbb{C}^{10} \rightarrow \mathbb{C}^2$ (Theorem 10.2). The separation of responsibilities is therefore

$$\text{ProjectionSVD: } (\alpha_\Pi, r, V_5, K_\Pi, H_\Pi), \quad \text{FMT: } (\mathcal{F}_\varphi, U_\varphi, \mathcal{R}_\chi).$$

Any physical or gauge interpretation of the resulting two-dimensional quotient belongs to downstream work and is not undertaken here.

2 The FMT and the Golden Lattice

Definition 2.1 (Golden fine lattice). $x_k = \varphi^{k/5}$, $k = 0, \dots, 9$; log-space steps $u_k = (k/5) \log \varphi$; quasi-periodic closure $x_{10} = \varphi^2 \neq 1$.

Definition 2.2 (Fibonacci–Mellin Transform).

$$(\mathcal{F}_\varphi)_{jk} = \frac{1}{\sqrt{10}} e^{i\omega_j u_k}, \quad \omega_j = \varphi^{(j-5)/5}, \quad j, k = 0, \dots, 9. \quad (2)$$

The 10 Mellin frequencies ω_j form a geometric sequence with ratio $\varphi^{1/5}$, centred at $\omega_5 = 1$. Writing $z_j = e^{i\theta_j}$ with $\theta_j = \omega_j \log \varphi/5$, $\mathcal{F}_\varphi$ is a Vandermonde matrix $(\mathcal{F}_\varphi)_{jk} = z_j^k/\sqrt{10}$; the 10 nodes z_j lie in an arc of 4.70° on S^1 .

Table 2: Structural comparison: FMT vs DFT.

Property	DFT on $\mathbb{Z}_{10}$	FMT on $\{x_k = \varphi^{k/5}\}$
Lattice	additive $\mathbb{Z}_{10}$	multiplicative $\{\varphi^{k/5}\}$
Closure	$\omega^{10} = 1$	$x_{10} = \varphi^2 \neq 1$
Kernel	$e^{-2\pi ijk/N}$ (Fourier)	$e^{i\omega_j u_k}$ (Mellin)
Indexing graph	$\mathbb{Z}_{10}$ (cycle)	GP(5,2), D_5 -symmetric (not a Cayley graph)
Output structure	N scalar bins $X_j \in \mathbb{C}$	5 doublets $\Psi_j \in \mathbb{C}^2$
Representation	scalar $U(1)$	$U(1) \times SU(2)$ -type
Twiddle phase	$2\pi k/N \in \mathbb{Q}\pi$	$\delta_{j,k}/\pi$: no rational commensurability detected
Symmetry	cyclic $\mathbb{Z}_{10}$	dihedral D_5 action; sheet-coupling twist
Role	frequency memory	scale memory

Note. The rows compare two finite constructions. The additive/multiplicative contrast is not an inequivalence of the underlying continuous structures: $u = \log x$ carries the multiplicative half-line onto the additive line unitarily, and the Mellin transform is the Fourier transform read in that coordinate (Remark 2.4). What separates the two finite transforms is the closure, the node geometry, and the resulting frame bounds.

Proposition 2.3 (Isometry deficit). *In exact arithmetic, evaluated in multiprecision and stable from 40 to 120 decimal digits (§9.1),*

$$\sigma_{\max}(\mathcal{F}_\varphi) = 3.153410735732771\dots, \quad \sigma_{\min}(\mathcal{F}_\varphi) = 6.70707500649 \times 10^{-19},$$

$$\kappa(\mathcal{F}_\varphi) = 4.70161841441 \times 10^{18}, \quad \|\mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\|_F = 9.42795.$$

Double precision does not resolve the last singular values: it returns $\sigma_{\min}$ of order 10^{-17} and κ of order 10^{17} , one to two orders of magnitude away from the certified values. The precise figures are not a property of the frame — they depend on how the matrix is built and on the linear-algebra library used, and they differ between machines (§9.1).

The 4.70° phase concentration and the isometry deficit are the same geometric fact: the Vandermonde nodes are nearly collinear on S^1 , so the FMT frame cannot distribute energy uniformly in the Euclidean metric.

Remark 2.4 (Where the non-unitarity comes from). The deficit of Proposition 2.3 belongs to the finite construction; it is not inherited from the Mellin transform. At the continuous level the map $(\mathcal{W}f)(u) = f(e^u)$ satisfies

$$\|\mathcal{W}f\|_{L^2(\mathbb{R}, du)}^2 = \int_{\mathbb{R}} |f(e^u)|^2 du = \int_0^\infty |f(x)|^2 \frac{dx}{x} = \|f\|_{L^2(\mathbb{R}_+, dx/x)}^2,$$

so $\mathcal{W} : L^2(\mathbb{R}_+, dx/x) \rightarrow L^2(\mathbb{R}, du)$ is unitary; it conjugates the self-adjoint dilation generator $-ix d/dx$ into $-i d/du$, and the Mellin–Plancherel transform factorises as

$\mathcal{M} = \mathcal{F}_c \mathcal{W}$, where $\mathcal{F}_c$ is the Fourier–Plancherel transform on $L^2(\mathbb{R}, du)$; it is therefore unitary. The three sources of the FMT deficit are instead all visible in Definition 2.2: the sampling is finite ($N = 10$ nodes), the frequencies ω_j form a geometric rather than an equispaced sequence, and the resulting Vandermonde nodes are non-orthogonal, concentrated in an arc of 4.70° . Multiplicative structure by itself is not the source of the deficit: finite, non-orthogonal sampling in the logarithmic coordinate is.

3 LOW/HIGH Two-Sheet Decomposition

3.1 Mellin strand and Artin twist

The Mellin frequencies satisfy the scale-doubling relation $\omega_{j+5} = \varphi \omega_j$ for $j = 0, \dots, 4$. This separates the 10 frequencies into two sheets of 5. Define, for $(j, k) \in \{0, \dots, 4\} \times \{0, \dots, 9\}$:

$$R_j(k) = e^{i\omega_j u_k} \quad (\text{Mellin strand}), \quad A_j(k) = e^{i\omega_j \varphi^{-1} u_k} \quad (\text{Artin twist}). \quad (3)$$

The Fibonacci identity $1 + \varphi^{-1} = \varphi$ (from $\varphi^2 = \varphi + 1$) gives the fundamental relation:

$$\boxed{A_j(k) \cdot R_j(k) = e^{i\omega_j(1+\varphi^{-1})u_k} = e^{i\omega_j \varphi u_k} = R_{j+5}(k)}. \quad (4)$$

The HIGH strand R_{j+5} is the LOW strand R_j twisted by A_j . This is the algebraic core of the entire paper.

Theorem 3.1 (LOW/HIGH two-sheet decomposition). *For $f \in \mathbb{C}^{10}$ and $y = \mathcal{F}_\varphi f$, define*

$$\text{LOW}_j = \frac{1}{\sqrt{10}} \sum_{k=0}^9 f[k] R_j(k) = y_j, \quad (5)$$

$$\text{HIGH}_j = \frac{1}{\sqrt{10}} \sum_{k=0}^9 f[k] A_j(k) R_j(k) = y_{j+5}. \quad (6)$$

The raw FMT decomposes into a LOW sheet $(y_0, \dots, y_4)$ and a HIGH sheet $(y_5, \dots, y_9)$, related entry-wise by

$$(\mathcal{F}_\varphi)_{j+5,k} = A_j(k) (\mathcal{F}_\varphi)_{j,k}, \quad (7)$$

so that HIGH_j is the φ -scaled replica of LOW_j via the Artin twist.

Proof. $(\mathcal{F}_\varphi)_{jk} = R_j(k)/\sqrt{10}$ by definition. By (4), $R_{j+5}(k) = A_j(k)R_j(k)$, hence $(\mathcal{F}_\varphi)_{j+5,k} = A_j(k)(\mathcal{F}_\varphi)_{jk}$ entry-wise. Summing over k : $y_{j+5} = \frac{1}{\sqrt{10}} \sum_k f[k] A_j(k) R_j(k) = \text{HIGH}_j$. The Artin twist acts *inside* the sum, not on the already-projected coefficient LOW_j ; the two sheets are related at the matrix-entry level, not as a scalar multiplication of the output coefficients. $\square$

Remark 3.2 (CT(2,2) is optional post-processing). Applying the Hadamard to the pair (y_j, y_{j+5}) produces *new combination channels*:

$$c_j^+ = y_j + y_{j+5}, \quad c_j^- = y_j - y_{j+5}. \quad (8)$$

These symmetric/antisymmetric combinations are not the raw FMT coefficients; they are an optional two-channel readout. The symmetric/antisymmetric channels provide an optional basis change within each LOW/HIGH pair. They are not part of the raw FMT and no additional dynamics, periodicity, or physical interpretation is inferred from this basis change.

3.2 GP(5,2) as graph model of the two-sheet indexing

Remark 3.3 (GP(5,2) encodes the sheet structure). The generalized Petersen graph GP(5,2) is the Petersen graph: it carries a faithful D_5 automorphism action and is vertex-transitive, but it is *not* a Cayley graph of D_5 — indeed it is the smallest vertex-transitive graph that is not a Cayley graph of any group. We use it here purely as a D_5 -symmetric *indexing graph* for the two sheets, with the outer/inner pentagons and radial edges as labels; no group multiplication is read off the edges. In the FMT:

- *Outer pentagon* (LOW sheet): the 5 LOW-sheet Mellin frequencies $\omega_0, \dots, \omega_4$; edges represent the monotone frequency progression $\omega_j \rightarrow \omega_{j+1} = \varphi^{1/5} \omega_j$.
- *Inner pentagon* (HIGH sheet): the 5 HIGH-sheet frequencies $\omega_5, \dots, \omega_9 = \varphi \omega_0, \dots, \varphi \omega_4$, traversed with the opposite orientation under the order-two symmetry.
- *Radial edges* (Petersen skip $\sigma = (0, 2, 4, 1, 3)$): the sheet coupling; each radial edge corresponds to an Artin twist $A_j(k)$.

The *abelian limit* $A_j \equiv 1$ collapses the two pentagons: HIGH = LOW and the two-sheet separation vanishes. The non-trivial Artin twist on the radial edges is what makes the two sheets distinct.

Figure 1 shows the 5×2 butterfly; Figure 2 shows the GP(5,2) graph.

4 Representation Structure: from Scalar Bins to Sheet Doublets

4.1 DFT: scalar $U(1)$ bins

Each DFT coefficient $X_j \in \mathbb{C}$ is a scalar: it carries amplitude $|X_j|$ and phase $\arg X_j$. The only internal symmetry is the $U(1)$ phase rotation $X_j \mapsto e^{i\theta} X_j$.

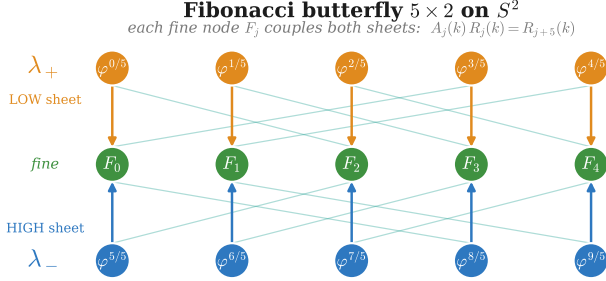


Figure 1: Fibonacci butterfly 5×2 on S^2 . Nodes $\varphi^{k/5}$ split into λ_+ (orange, LOW, $k = 0 \dots 4$) and λ_- (blue, HIGH, $k = 5 \dots 9$). Crossing arrows from both sheets to each fine node F_j represent the Artin twist coupling $A_j(k)R_j(k) = R_{j+5}(k)$.

GP(5,2): D_5 -symmetric indexing graph
(the Petersen graph — not a Cayley graph)

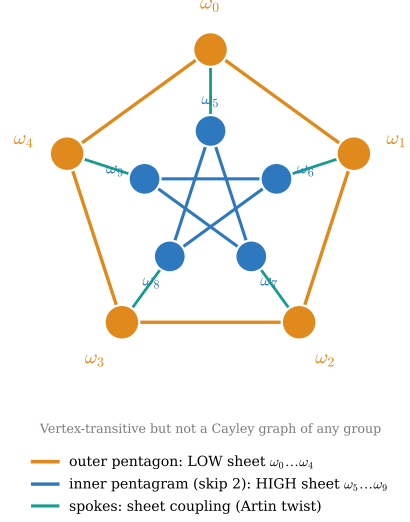


Figure 2: D_5 -symmetric indexing graph GP(5,2) (the Petersen graph, not a Cayley graph). Outer pentagon (orange): LOW-sheet frequencies. Inner pentagon (blue): HIGH-sheet frequencies (opposite orientation). Radial edges (teal): sheet-coupling Artin twist.

4.2 FMT: doublet with $U(2)/SU(2)$ structure

Group the two FMT sheets. At each frequency index $j = 0, \dots, 4$ and each sample k , define the *local* scale doublet

$$\psi_j(k) = \begin{pmatrix} R_j(k) \\ A_j(k) R_j(k) \end{pmatrix} \in \mathbb{C}^2, \quad (9)$$

on which the index-dependent φ -memory cell $G_j(k) \in U(2)$ acts pointwise, *before* the projection and the sum over k . The global doublet is the post-summation object

$$\Psi_j = \begin{pmatrix} \text{LOW}_j \\ \text{HIGH}_j \end{pmatrix} = \frac{1}{\sqrt{10}} \sum_k f[k] \psi_j(k) = \begin{pmatrix} y_j \\ y_{j+5} \end{pmatrix} \in \mathbb{C}^2. \quad (10)$$

Because $G_j(k)$ depends on k , it acts on the local doublet $\psi_j(k)$, not pointwise on Ψ_j ; after removing the global phase (see §6), $\tilde{G}_j(k) \in SU(2)$.

Proposition 4.1 ($U(1) \times SU(2)$ -type representation). *The DFT resolves scalar $U(1)$ characters. The FMT resolves Mellin $U(1)$ characters together with a two-sheet internal fibre: grouping the sheets into the local doublet $\psi_j(k)$, the local phase cell*

$G_j(k) \in U(2)$ acts; after removal of the global phase $e^{-i(\delta_{j,k}+\pi)/2}$, the reduced cell $\tilde{G}_j(k) \in SU(2)$ satisfies $\det \tilde{G}_j(k) = 1$. In this representation-theoretic sense, the FMT exposes a $U(1) \times SU(2)$ -type readout structure that is not present in the canonical scalar-bin DFT readout unless an additional pairing of bins is imposed by hand. We stress that this is a structure of the LOW/HIGH readout induced by the two-sheet decomposition, not a new physical symmetry.

Proof. From (18): $\det G_j(k) = -e^{i\delta_{j,k}}$, so $|\det G_j| = 1$ and $G_j \in U(2)$. Setting $\tilde{G}_j = e^{-i(\delta_{j,k}+\pi)/2} G_j$ gives $\det \tilde{G}_j = e^{-i(\delta_{j,k}+\pi)} \cdot (-e^{i\delta_{j,k}}) = 1$, hence $\tilde{G}_j \in SU(2)$. Verification: $\|\tilde{G}_j^* \tilde{G}_j - I_2\|_F < 4 \times 10^{-16}$. $\square$

4.3 Stokes-like observables on the doublet

The doublet Ψ_j carries four real observables, analogous to Stokes parameters for a two-component field:

$$S_0(j) = |\text{LOW}_j|^2 + |\text{HIGH}_j|^2 \quad (\text{total scale energy}), \quad (11)$$

$$S_x(j) = 2 \operatorname{Re}(\text{LOW}_j^* \text{HIGH}_j) \quad (\text{in-phase cross-sheet coherence}), \quad (12)$$

$$S_y(j) = 2 \operatorname{Im}(\text{LOW}_j^* \text{HIGH}_j) \quad (\text{quadrature cross-sheet coherence}), \quad (13)$$

$$S_z(j) = |\text{LOW}_j|^2 - |\text{HIGH}_j|^2 \quad (\text{sheet energy imbalance}). \quad (14)$$

These satisfy $S_x^2 + S_y^2 + S_z^2 = S_0^2$ (verified numerically with error $< 10^{-14}$), so the doublet lies on the Poincaré sphere of radius $S_0(j)$.

Complementary observables extracted from the sheet ratio:

$$\rho_j = \frac{|\text{HIGH}_j|}{|\text{LOW}_j|}, \quad \psi_j = \angle\left(\frac{\text{HIGH}_j}{\text{LOW}_j}\right), \quad \text{LOW}_j \neq 0, \quad (15)$$

Here ψ_j is the effective inter-sheet phase induced by the pointwise Artin-twisted projection: since $\text{HIGH}_j = \frac{1}{\sqrt{10}} \sum_k f[k] A_j(k) R_j(k)$ with $A_j(k) = e^{i\delta_{j,k}}$ acting inside the sum, ψ_j encodes the signal-weighted average of $\delta_{j,k}$ over the golden lattice. For a φ -rational signal the dominant lattice contributions drive ψ_j toward the golden progression $\delta_{j,k}$; for a harmonic signal ψ_j reflects the harmonic spectral texture.

Candidate domains. For OMA/structural health monitoring: damage may be modeled as a perturbation of the cross-sheet observables (S_x, S_y, S_z) ; whether these features improve over MAC criterion, SSI, PCA, or Mahalanobis indicators remains an open validation problem (see Table 1). For SAR/remote sensing: the sheet imbalance S_z measures the energy asymmetry between scale levels, complementing amplitude sparsity (Gini) with a phase-internal scale feature; systematic benchmarking against standard texture descriptors is required.

FMT DSP cell: GP(5,2) core $\circ$ CT(2,2) border

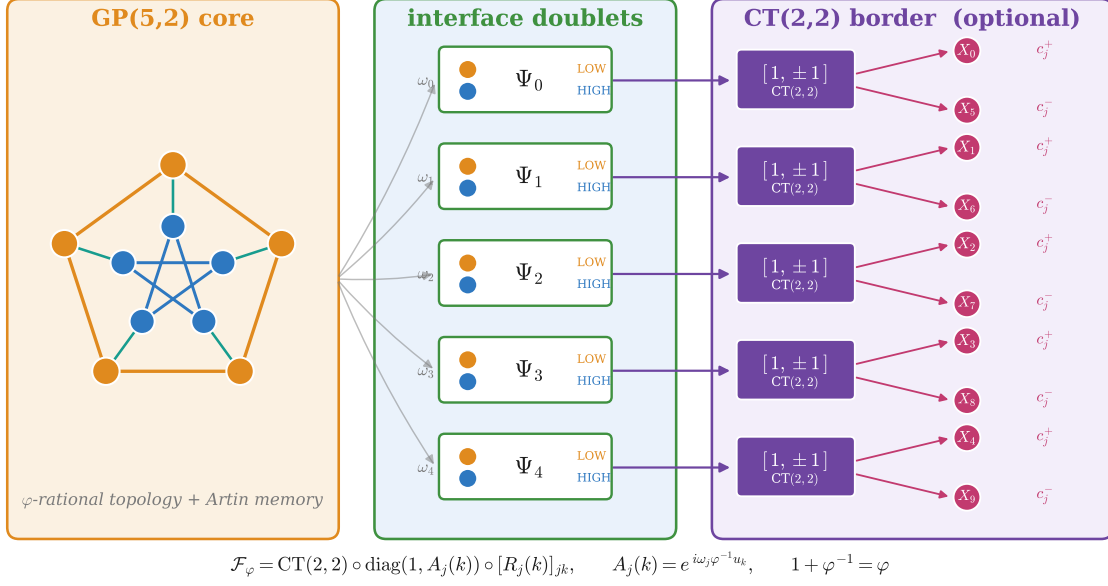


Figure 3: DSP cell structure. **Left:** GP(5,2) core — outer pentagon LOW (orange), inner pentagon HIGH (blue), radial edges = Artin twist A_j (teal). **Centre:** 5 interface doublets $\Psi_j = (\text{LOW}_j, \text{HIGH}_j)^T$. **Right:** optional CT(2,2) post-processing creating symmetric/antisymmetric channels $c_j^\pm = y_j \pm y_{j+5}$.

5 Short-Time FMT and the Scale-Transfer Spectrum

The representation structure of §4 is local: it acts on one ten-node cell. To analyse a signal of arbitrary length we window it into successive cells and recurse, in exact analogy with the short-time Fourier transform (STFT). We call the result the *Short-Time Fibonacci–Mellin Transform* (STFMT) and derive its natural output observable.

5.1 Windowing and the recursive golden filter bank

Let $f \in \mathbb{C}^L$ be sampled on the golden log-lattice. Partition it into non-overlapping (or hop-spaced) blocks of $N = 2R = 10$ nodes; apply the frame $\mathcal{F}_\varphi$ to each block, yielding for block m the two sheets $\text{LOW}^{(m)}, \text{HIGH}^{(m)} \in \mathbb{C}^R$ as in (10). The STFMT is the indexed family $\{(\text{LOW}^{(m)}, \text{HIGH}^{(m)})\}_m$.

Because the lattice is multiplicatively self-similar, we define a recursive golden-scale construction by reapplying the ten-node FMT to successive blocks on the logarithmic axis. This is an algorithmic extension of the finite frame. No unique multiresolution analysis, orthogonal filter bank, or dedicated fast transform is claimed. Fast Mellin/scale algorithms exist for related scale-transform settings [7, 8], but no dedicated fast algorithm for the golden FMT is proved here.

5.2 The SVD-inherited Pincherle–Mellin transfer representative

The upstream projection–SVD construction [1] supplies the rank-five logarithmic block $V_5 = \text{span}\{1, u, u^2, u^3, u^4\}$, $u = \log x$, the quadratic Pincherle threshold $\alpha_\Pi = \frac{1}{20\varphi^4} = (7 - 3\sqrt{5})/40$, and an auxiliary rank-five rational transfer representative

$$H_\Pi(z) = \frac{P_4(z)}{1 - \alpha_\Pi z}, \quad P_4(z) = \sum_{n=0}^4 (\log \varphi)^n z^n. \quad (16)$$

These objects are imported; their Pincherle derivation and their role in selecting $r = 5$ are not repeated here. Like its SVD counterpart, H_Π is an auxiliary filter associated with the same scale α_Π and the same rank-five truncation degree; it is *not* the boundary metric, and no identity $H_\Pi \leftrightarrow K_\Pi$ is claimed.

The FMT contribution is the finite Mellin realization of this logarithmic structure. Since $\mathcal{D} = x d/dx = d/du$, the Mellin characters $m_\omega(x) := x^{i\omega} = e^{i\omega u}$ satisfy $\mathcal{D}m_\omega = i\omega m_\omega$. Sampling them at $u_k = (k/5) \log \varphi$, $\omega_j = \varphi^{(j-5)/5}$ gives exactly $(\mathcal{F}_\varphi)_{jk} = \frac{1}{\sqrt{10}} m_{\omega_j}(x_k) = \frac{1}{\sqrt{10}} e^{i\omega_j u_k}$, so $\mathcal{F}_\varphi$ is the finite evaluation matrix of the Mellin eigencharacters of the Pincherle generator on the golden lattice.

The numerator of H_Π is a truncated geometric polynomial, $P_4(z) = (1 - ((\log \varphi)z)^5)/(1 - (\log \varphi)z)$, with four non-trivial zeros $z_k = (\log \varphi)^{-1} e^{2\pi i k/5}$, $k = 1, 2, 3, 4$. This is a fivefold spectral zero geometry inherited from the rank-five block; it does not imply a five-step-periodic impulse response. In the causal DSP convention $H_{\Pi,c}(z) = P_4(z^{-1})/(1 - \alpha_\Pi z^{-1})$ the single pole is at $z = \alpha_\Pi$, inside the unit disk, with exact recursion $y_k = \sum_{n=0}^4 (\log \varphi)^n u_{k-n} + \alpha_\Pi y_{k-1}$: a five-tap finite numerator followed by one-pole memory. No physical oscillation, resonant mode, or dimensionless frequency is deduced from the zero pattern.

5.3 The natural invariant: scale-transfer, not power

The STFT is paired with the power spectral density (PSD): Fourier is a *unitary*, *additive* decomposition, Parseval holds, and the conserved per-bin quantity is the energy $|X_j|^2$. The FMT differs on both counts. It is *non-unitary* ($\|\mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\|_F \approx 9.4$, the projection signature of §8) — a property of the finite, non-orthogonal sampling and not of the Mellin transform, which is unitary (Remark 2.4) — so Parseval fails and $|y_j|^2$ is *not* conserved; and its split is *multiplicative* (two scale sheets) rather than additive (DC vs. Nyquist). The quantity invariant under the operative symmetry — global gain $f \mapsto \lambda f$ — is therefore not the additive energy but the *multiplicative* sheet ratio. No dynamical evolution is introduced here, so *invariance* under a group action is meant throughout, never a conservation law in time. We define the **scale-transfer**

spectrum

$$T_j = \log \frac{\text{HIGH}_j}{\text{LOW}_j} = \underbrace{\log \left| \frac{\text{HIGH}_j}{\text{LOW}_j} \right|}_{\ell_j} + i \underbrace{\text{unwrap arg } \frac{\text{HIGH}_j}{\text{LOW}_j}}_{\psi_j} \quad j = 0, \dots, R-1, \quad (17)$$

the complex logarithm of the doublet ratio of (10), written in stable log-polar form (magnitude ℓ_j , unwrapped phase ψ_j) to avoid branch-cut artefacts. The transform T_j is defined on the domain $\text{LOW}_j \text{ HIGH}_j \neq 0$; for the numerical algorithm, sub-threshold bins are regularised, e.g.

$$T_j^{(\varepsilon)} = \log \frac{\text{HIGH}_j + \varepsilon e^{i \arg \text{HIGH}_j}}{\text{LOW}_j + \varepsilon e^{i \arg \text{LOW}_j}},$$

or bins with $|\text{LOW}_j|$ or $|\text{HIGH}_j|$ below a fixed threshold are masked. The correspondence is exact:

	DFT butterfly 2×2	FMT butterfly 5×2
split	additive ($a \pm b$)	multiplicative (LOW, HIGH)
unitary	yes (Parseval)	no ($\ \mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\ _F \approx 9.4$)
invariant	energy $ X ^2$	log-ratio $\log(\text{HIGH}/\text{LOW})$
readout	power spectrum	scale-transfer spectrum

Gain invariance. Since T_j is a ratio of two co-scaled linear projections, it is *exactly* invariant under global complex gain, $T_j(\lambda f) = T_j(f)$ for all $\lambda \in \mathbb{C}^*$ — the same algebraic mechanism that underlies a validation gate investigated in downstream applied work. The magnitude ℓ_j measures how much structure is transferred between adjacent golden scales; the phase ψ_j measures how the scale phase rotates. PSD is flat or gain-dependent on this signal class and does not isolate this quantity.

5.4 What the scale-transfer spectrum is and is not

The scale-transfer spectrum is a *readout of scale structure*, not a competitor to Fourier or wavelet analysis on their own terrain. Three limits should be stated plainly, each verified empirically:

- (i) **Not a faster transform.** The golden recursion inherits no dedicated fast algorithm; fixed-size cells are used, and fast Mellin/scale algorithms for related settings are not specialized to the golden FMT here.
- (ii) **Not a better compressor.** The frame does not preserve the Euclidean norm or Parseval energy; depending on the input direction it may amplify or attenuate the Euclidean norm. An orthogonal wavelet bank reconstructs a signal more accurately at a fixed coefficient budget. The STFMT nonetheless improves on the single-pass (non-recursive) FMT, confirming that the recursive bank adds representational structure.

- (iii) **Its value is invariance, not discrimination.** The gain-invariance of T_j is exact and is the basis of its applied use as a validation gate in downstream applied work; we do not claim that T_j out-discriminates standard descriptors as a primary feature.

In short, the STFMT supplies a parameter-free, gain-invariant scale readout whose engineering value is stability under the acquisition chain, developed quantitatively in the applied companion paper.

6 The φ -Memory Cell

6.1 Definition and unitarity

Proposition 6.1 (φ -memory cell). *The 2×2 operator at each interface node F_j is*

$$G_j(k) = H \cdot \text{diag}(1, e^{i\delta_{j,k}}), \quad \delta_{j,k} = u_k \omega_j \varphi^{-1} = \frac{k}{5} \log \varphi \cdot \varphi^{(j-10)/5}, \quad (18)$$

where $H = [\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}]/\sqrt{2}$. $G_j(k) \in U(2)$ for all (j, k) . The phase $\delta_{j,k}$ depends on lattice index k and frequency index j , not on the signal $f[k]$.

$$\text{Explicitly: } G_j(k) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & e^{i\delta_{j,k}} \\ 1 & -e^{i\delta_{j,k}} \end{bmatrix}.$$

6.2 Phase structure and quasi-periodicity

$$\delta_{j,k} \in \mathbb{R}_{>0} \cdot \log \varphi, \quad \text{with algebraic coefficient in the fine-lattice field } \mathbb{Q}(\varphi^{1/5}). \quad (19)$$

Explicitly, $\delta_{j,k} = (k/5) \log \varphi \cdot \varphi^{(j-10)/5}$, where $\varphi^{(j-10)/5} \in \mathbb{Q}(\varphi^{1/5})$ for $j = 0, \dots, 4$ (not in $\mathbb{Q}(\sqrt{5}) = \mathbb{Q}(\varphi)$ unless $j \equiv 0 \pmod{5}$). Whether $\delta_{j,k}/\pi$ is irrational for all (j, k) is a transcendence question that we do not resolve here. No exact irrationality or infinite-order theorem is claimed. Numerical tests on the stated lattice detected no return to the identity within the tested exponent range and tolerance; only the qualitative statement that no finite order was detected in the reported numerical experiment is retained.

6.3 Relation to the Fibonacci anyon braid matrices

The Fibonacci anyon $\sigma_1 = \text{diag}(e^{-4\pi i/5}, e^{3\pi i/5})$ has phases in $\mathbb{Q}\pi$ and finite order 10. $G_j(k)$ has phase in $\mathbb{R}_{>0} \cdot \log \varphi$ with algebraic coefficient in $\mathbb{Q}(\varphi^{1/5})$, and no detected finite order (see above). The two objects are complementary: σ_1, σ_2 implement *topological braiding* (discrete phase, finite order, exact algebraic computation); $G_j(k)$ implements *spectral scale memory* (continuous quasi-periodic phase, golden-lattice geometry). The same pentagonal/dihedral combinatorics of GP(5,2) appears in graphical descriptions of Fibonacci-type braid structures [4], but the present paper uses GP(5,2) only as an indexing graph for the FMT two-sheet decomposition.

Table 3: The φ -memory cell vs standard operators. $G_j(k)$ is index-dependent; all others are fixed.

Operator	Phase δ	$\delta/\pi \in \mathbb{Q}$?	Finite order?	Index-dep.?
Hadamard H	0	yes	$H^2 = I$	no
T gate	$\pi/4$	yes	order 8	no
S gate	$\pi/2$	yes	order 4	no
σ_1 (Fib. anyon)	$\{-4, 3\}\pi/5$	yes	order 10	no
$G_j(k)$ (φ -memory)	$u_k \omega_j \varphi^{-1}$, 4–28°	not detected	none detected	yes

7 Scale-Coherence Analysis Algorithm

The following algorithm converts a signal $f[k]$ on the golden lattice into a complete scale-coherence feature vector.

Algorithm 1 FMT Scale-Coherence Analysis

Require: Signal $f \in \mathbb{C}^{10}$ on golden lattice $x_k = \varphi^{k/5}$, DFT matrix $D \in \mathbb{C}^{10 \times 10}$

Ensure: Scale-coherence feature vector $\mathbf{v}$

- 1: Compute raw FMT: $y \leftarrow \mathcal{F}_\varphi f$, DFT: $\hat{f} \leftarrow Df$
- 2: Extract sheets: $\text{LOW}_j \leftarrow y_j$, $\text{HIGH}_j \leftarrow y_{j+5}$, $j = 0, \dots, 4$
- 3: Form doublets: $\Psi_j \leftarrow (\text{LOW}_j, \text{HIGH}_j)^T$
- 4: Compute Stokes observables per $j = 0, \dots, 4$:

$$S_0(j) \leftarrow |\text{LOW}_j|^2 + |\text{HIGH}_j|^2$$

$$S_x(j) \leftarrow 2 \text{Re}(\text{LOW}_j^* \text{HIGH}_j), \quad S_y(j) \leftarrow 2 \text{Im}(\text{LOW}_j^* \text{HIGH}_j), \quad S_z(j) \leftarrow |\text{LOW}_j|^2 - |\text{HIGH}_j|^2$$

- 5: Compute Artin phase (FMT and DFT):

$$\Delta\varphi_j^{\text{FMT}} \leftarrow \arg(y_{j+1}/y_j), \quad j = 0, \dots, 8$$

$$\Delta\varphi_j^{\text{DFT}} \leftarrow \arg(\hat{f}_{j+1}/\hat{f}_j), \quad j = 0, \dots, 8$$

- 6: Fit against golden target $\tau_j = \varphi^{j/5}$ (affine least-squares):

$$R_{\text{FMT}}^2 \leftarrow R^2(\Delta\varphi^{\text{FMT}}, \tau), \quad \text{TV}_{\text{FMT}} \leftarrow \sum_j |\Delta\varphi_{j+1}^{\text{FMT}} - \Delta\varphi_j^{\text{FMT}}|$$

$$R_{\text{DFT}}^2 \leftarrow R^2(\Delta\varphi^{\text{DFT}}, \tau), \quad \text{TV}_{\text{DFT}} \leftarrow \sum_j |\Delta\varphi_{j+1}^{\text{DFT}} - \Delta\varphi_j^{\text{DFT}}|$$

- 7: Compute contrasts: $\Delta\text{TV} \leftarrow \text{TV}_{\text{DFT}} - \text{TV}_{\text{FMT}}$, $\Delta R^2 \leftarrow R_{\text{FMT}}^2 - R_{\text{DFT}}^2$
 - 8: **return** $\mathbf{v} = (S_0(j), S_x(j), S_y(j), S_z(j), \psi_j, \Delta\varphi_j^{\text{FMT}}, R_{\text{FMT}}^2, \text{TV}_{\text{FMT}}, \Delta\text{TV}, \Delta R^2)_{j=0}^4$
-

The output $\mathbf{v}$ has $5 \times 5 + 9 + 4 = 38$ scalar components: 20 Stokes observables (S_0, S_x, S_y, S_z) for $j = 0, \dots, 4$; 5 inter-sheet phases $\psi_j = \angle(\text{HIGH}_j/\text{LOW}_j)$; 9 Artin phase increments $\Delta\varphi_j^{\text{FMT}}$; and 4 global metrics ($R_{\text{FMT}}^2, \text{TV}_{\text{FMT}}, \Delta\text{TV}, \Delta R^2$). For practical classification tasks, the most discriminating subset is $(S_z(j), \Delta R^2, \Delta\text{TV}, \psi_j)$. *Primary separator*: ΔR^2 separates φ -structured signals ($\Delta R^2 > 0.5$) from harmonic controls ($\Delta R^2 \approx 0.2$) across all tested classes. *Secondary descriptor*: ΔTV should not be used alone, since harmonic controls can produce large DFT phase variation ($\Delta\text{TV}_{\text{harmonic}} \approx 13$) comparable to some φ -structured cases; it is useful as a complementary feature after thresholding on ΔR^2 .

8 Diagonal Unitarization Obstruction

Proposition 8.1 (No diagonal unitarization — numerically certified for $N = 10$). *There is no $(w_0, \dots, w_9) \in \mathbb{R}^{10}$ such that $\mathcal{F}_\varphi \text{diag}(w_0, \dots, w_9) \mathcal{F}_\varphi^* = I_{10}$.*

Proof. Write $P_w(z) = \sum_{k=0}^9 w_k z^k$, $\deg P_w \leq 9$.

Diagonal conditions. $(\mathcal{F}_\varphi W \mathcal{F}_\varphi^*)_{jj} = 1$ gives $\frac{1}{10} \sum_k w_k = 1$, i.e. $P_w(1) = 10$.

Off-diagonal conditions. $(\mathcal{F}_\varphi W \mathcal{F}_\varphi^*)_{jj'} = 0$ for $j \neq j'$ requires $P_w(z_{jj'}) = 0$, where $z_{jj'} = e^{i(\theta_j - \theta_{j'})}$, $\theta_j = \omega_j \log \varphi / 5$. Since $\theta_j \neq \theta_{j'}$ for $j \neq j'$, the point $z = 1$ is *not* among the 90 off-diagonal values.

Distinctness (double-precision numerical certification). All 90 values $z_{jj'} \bmod 2\pi$ are pairwise distinct: $\min_{(j,j') \neq (m,m')} |z_{jj'} - z_{mm'}| \geq 3.5 \times 10^{-4}$, far above double-precision rounding ($\sim 10^{-16}$). This also disposes of the $2\pi n$ ambiguity: differences are computed and compared modulo 2π directly.

Contradiction. $\deg P_w \leq 9$ implies P_w has at most 9 complex roots. It cannot vanish at 90 distinct points, so $P_w \equiv 0$, contradicting $P_w(1) = 10$. $\square$

Corollary 8.2. *The isometry deficit $\|\mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\|_F = 9.43$ is not reducible by any real diagonal reweighting (Proposition 8.1).*

Remark 8.3 (Modular orthogonality). Exact orthogonality holds over $\mathbb{F}_{11}$ (where 2 has multiplicative order $10 = 2r$) and via a 13-point arithmetic mask [1].

9 Polar Unitary Readout

Remark 9.1 (The polar unitary factor). The factor $U_\varphi = WV^*$, obtained from the SVD $\mathcal{F}_\varphi = W\Sigma V^*$, is the *polar unitary factor* of the frame: the closest unitary to $\mathcal{F}_\varphi$ in Frobenius norm. It is a classical linear-algebra object. The embedding into $U(16)$ and the block-encoding below are stated only to characterize this unitary readout; no quantum-circuit implementation or qubit realization is claimed. Readers interested only in the DSP content may treat this section as defining a unitary alternative readout and skip the embedding details.

Theorem 9.2 (Polar unitary readout). *Let $\mathcal{F}_\varphi = W\Sigma V^*$ be the SVD. Define $U_\varphi = WV^*$. Then: $U_\varphi \in U(10)$, $U_\varphi = \mathcal{F}_\varphi(\mathcal{F}_\varphi^*\mathcal{F}_\varphi)^{-1/2}$ in exact arithmetic, and $\|U_\varphi^*U_\varphi - I_{10}\|_F < 3 \times 10^{-15}$.*

Proof. $\mathcal{F}_\varphi(\mathcal{F}_\varphi^*\mathcal{F}_\varphi)^{-1/2} = W\Sigma V^* \cdot V\Sigma^{-1}V^* = WV^*$; $U_\varphi^*U_\varphi = VW^*WV^* = I$. The SVD route avoids the ill-conditioned direct computation of $(\mathcal{F}_\varphi^*\mathcal{F}_\varphi)^{-1/2}$. $\square$

Remark 9.3. Because $|A_j(k)| = 1$ for all (j, k) , each sheet strand is already unimodular. $U_\varphi = WV^*$ is the global unitary completion that coherently assembles all 5 doublets Ψ_j into a single $U(10)$ operator; it is the natural reversible readout associated with the FMT frame. The polar factor can also be embedded into $U(16)$ via $U_\varphi \oplus I_6$, although circuit synthesis and quantum implementation are outside the scope of this paper.

Table 4: Matrix diagnostics.

Quantity	DFT	$\mathcal{F}_\varphi$	U_φ
Exact rank	10	10	10
Effective numerical rank	10	5 at tol 10^{-6} (6 at 10^{-8})	10
κ	1	4.70×10^{18}	1.00
$\sigma_{\max}$	1	3.1534	1
$\ A^*A - I\ _F$	0	9.43	2.8×10^{-15}

9.1 Certified numerics and the limits of double precision

Every number quoted for $\mathcal{F}_\varphi$ in this paper is produced in multiprecision arithmetic by the accompanying script `certify_precision.py`; double-precision values are reported only as comparison. The reason is the conditioning of the frame.

From the certified spectrum, the effective numerical rank at absolute tolerance τ is 4 for $\tau = 10^{-4}$, 5 for $\tau = 10^{-6}$, 6 for $\tau = 10^{-8}$ and 10^{-10} , and 7 for $\tau = 10^{-12}$: the frame has full exact rank 10 and a strongly graded spectrum, not a rank defect.

Accuracy of the specific polar factor. For a non-singular square matrix the unitary polar factor is unique, and its sensitivity is governed by the smallest singular value: for $B = A + E$,

$$\|U_p(B) - U_p(A)\|_F \leq \frac{2}{\sigma_{\min}(A) + \sigma_{\min}(B)} \|E\|_F, \quad (20)$$

a standard bound for the unitary factor [9]. Building $\mathcal{F}_\varphi$ in double precision perturbs it by $\|E\|_F \approx 4 \times 10^{-16}$, so (20) gives a bound of order 10^3 : at this conditioning double precision provides no control on U_φ at all. This is measured, not feared, and the

Table 5: Singular extrema of $\mathcal{F}_\varphi$: multiprecision certification against double precision. The multiprecision values are identical at every listed working precision.

Arithmetic	$\sigma_{\min}(\mathcal{F}_\varphi)$	$\kappa(\mathcal{F}_\varphi)$
40, 60, 80, 100, 120 digits	$6.70707500649 \times 10^{-19}$	$4.70161841441 \times 10^{18}$
double precision	$\sim 10^{-17}$	$\sim 10^{17}$

Note. The multiprecision row is exact and identical at every listed working precision. The double-precision row is deliberately given only by order of magnitude: across three constructions of the same matrix and two independent machines the observed values run from 5×10^{-18} to 2×10^{-17} for $\sigma_{\min}$ and from 1.5×10^{17} to 6×10^{17} for κ . These are artefacts of the arithmetic, not quantities of the frame, and are not reproducible across implementations.

measurement has a feature worth stating plainly: the deviation is $O(1)$ but its value is not reproducible. Against the 120-digit factor, the accompanying certifier records

$$0.5 \leq \|U_\varphi^{(64)} - U_\varphi^{(120)}\|_F \leq 3, \quad 0.05 \leq \|P_\chi^{(64)} - P_\chi^{(120)}\|_F \leq 1, \quad (21)$$

for every double-precision construction tested, on every machine tested; the individual numbers inside these ranges change with the construction of the matrix and with the linear-algebra library — three constructions on two independent hosts have produced values from 0.68 to 1.86 for U_φ and from 0.15 to 0.42 for P_χ . The same mathematical matrix, admissible floating-point constructions, polar factors differing at $O(1)$ and differing from each other: that is the statement, and pinning a particular value would misrepresent it. Restricted to the five leading right singular directions the two factors agree to $\leq 10^{-10}$: what double precision loses is the orientation of the ill-conditioned complement, not the retained block.

What this does and does not affect. The identities of Theorem 10.2 — $\mathcal{R}_\chi \mathcal{R}_\chi^* = I_2$, $P_\chi^2 = P_\chi$, $\text{rank } P_\chi = 2$, $\dim \ker \mathcal{R}_\chi = 8$ — hold for *any* unitary and are therefore satisfied by whichever factor is computed: the measured coisometry defect is $\approx 5 \times 10^{-16}$ in double precision and $\approx 10^{-120}$ at 120 digits. What double precision does not deliver is the identification of the computed unitary with the polar factor of the exact frame. Statements about the *specific* U_φ and P_χ of this frame are consequently certified only in multiprecision, and are so certified here.

Remark 9.4 (Embedding and POVM). The polar factor can be embedded into $U(16)$ via $U_\varphi \oplus I_6$, error $< 3 \times 10^{-15}$. Row projectors $E_j = |u_j\rangle\langle u_j|$ form a canonical POVM satisfying $\sum_j E_j = I_{10}$, error $< 3 \times 10^{-15}$. Halmos block-encoding gives $W_H \in U(20)$, $K = \mathcal{F}_\varphi / \sigma_{\max}$, error $< 9 \times 10^{-15}$. Circuit synthesis and quantum implementation are outside the scope of this paper.

10 Coherent Rank-One Reduction

The polar factor $U_\varphi \in U(10)$ of the previous section assembles the five doublets into a single reversible readout. We now record a second, exact, consequence of that readout: the ten-dimensional carrier reduces canonically to a two-dimensional coherent quotient. This reduction uses only the unitarity of U_φ and the dihedral symmetry of the register index; it is purely an operator-theoretic statement and assigns no physical meaning to the two retained components.

10.1 The five-register two-sheet carrier

The $N = 10$ index splits as a five-register frequency index $j = 0, \dots, 4$ times a two-valued sheet label (LOW/HIGH), so the carrier factorises as

$$\mathbb{C}^{10} = V_5 \otimes \mathbb{C}^2, \quad \dim_{\mathbb{C}} V_5 = 5, \quad \dim_{\mathbb{C}} \mathbb{C}^2 = 2, \quad (22)$$

where V_5 carries the five Mellin registers $\{|j\rangle\}_{j=0}^4$ and $\mathbb{C}^2$ carries the LOW/HIGH doublet $\Psi_j = (\text{LOW}_j, \text{HIGH}_j)^T$ of §4. The register shift $j \mapsto j + 1 \pmod{5}$ generates a cyclic $\mathbb{Z}_5$ action on V_5 , the rotation subgroup of the dihedral indexing symmetry D_5 of §3.

10.2 The dihedral invariant line

Lemma 10.1 (Unique D_5 -coherent line). *The uniform register superposition*

$$|\Omega\rangle = \frac{1}{\sqrt{5}} \sum_{j=0}^4 |j\rangle \in V_5, \quad \langle \Omega | \Omega \rangle = 1, \quad (23)$$

spans the unique line in V_5 fixed by the cyclic register action; its rank-one projector is $p_\Omega = |\Omega\rangle\langle\Omega|$.

Proof. The register shift acts on V_5 as the cyclic permutation matrix C with $C|j\rangle = |j + 1 \pmod{5}\rangle$. Its eigenvalues are the fifth roots of unity $e^{2\pi i m/5}$, $m = 0, \dots, 4$, each simple; the eigenvalue 1 has the one-dimensional eigenspace spanned by the all-ones vector $\sum_j |j\rangle$, normalised to $|\Omega\rangle$. The reflection generator of D_5 fixes this vector as well, so $|\Omega\rangle$ is the unique D_5 -invariant line. $\square$

10.3 The coisometric reduction

Theorem 10.2 (Coherent rank-one reduction $\mathbb{C}^{10} \rightarrow \mathbb{C}^2$). *Let $U_\varphi = WV^* \in U(10)$ be the polar factor of Theorem 9.2 and $|\Omega\rangle$ the D_5 -coherent line of Lemma 10.1. The coherent readout map*

$$\mathcal{R}_\chi := (\langle \Omega | \otimes I_2) U_\varphi : \mathbb{C}^{10} \longrightarrow \mathbb{C}^2 \quad (24)$$

satisfies

$$\mathcal{R}_\chi \mathcal{R}_\chi^* = I_2, \quad P_\chi := \mathcal{R}_\chi^* \mathcal{R}_\chi = U_\varphi^*(p_\Omega \otimes I_2) U_\varphi, \quad (25)$$

with $P_\chi^2 = P_\chi = P_\chi^*$, $\text{rank } P_\chi = 2$, and $\dim \ker \mathcal{R}_\chi = 8$. Hence the coherent observable quotient is canonically two-dimensional,

$$\mathbb{C}^{10} / \ker \mathcal{R}_\chi \simeq \mathbb{C}^2. \quad (26)$$

Proof. Since U_φ is unitary, $U_\varphi U_\varphi^* = I_{10}$, and since $\langle \Omega | \Omega \rangle = 1$,

$$\mathcal{R}_\chi \mathcal{R}_\chi^* = (\langle \Omega | \otimes I_2) U_\varphi U_\varphi^* (| \Omega \rangle \otimes I_2) = \langle \Omega | \Omega \rangle I_2 = I_2,$$

so $\mathcal{R}_\chi$ is a coisometry. Its adjoint gives $P_\chi = \mathcal{R}_\chi^* \mathcal{R}_\chi = U_\varphi^*(p_\Omega \otimes I_2) U_\varphi$, unitarily conjugate to the rank-two orthogonal projector $p_\Omega \otimes I_2$ ($p_\Omega^2 = p_\Omega = p_\Omega^*$, $\text{rank } p_\Omega = 1$, tensored with I_2). Therefore $P_\chi^2 = P_\chi = P_\chi^*$ and $\text{rank } P_\chi = 2$, whence $\dim \ker \mathcal{R}_\chi = 10 - 2 = 8$ and $\mathbb{C}^{10} / \ker \mathcal{R}_\chi \cong \text{Ran } \mathcal{R}_\chi = \mathbb{C}^2$. $\square$

Remark 10.3 (Status and compatibility). The reduction is *exact* and holds for any unitary U_φ and any unit register line; the FMT supplies the specific $U_\varphi = WV^*$ and the dihedral symmetry supplies the specific $|\Omega\rangle$. The identities $\mathcal{R}_\chi \mathcal{R}_\chi^* = I_2$, $\text{rank } P_\chi = 2$, $\dim \ker \mathcal{R}_\chi = 8$ were verified numerically on the $N = 10$ frame, in double precision ($\|\mathcal{R}_\chi \mathcal{R}_\chi^* - I_2\| < 10^{-15}$) and at 120 digits ($< 10^{-120}$). These identities hold for any unitary factor; the identification of the computed factor with the polar factor of the exact frame requires multiprecision and is certified in §9.1. The eight kernel directions are the part of the carrier annihilated by the coherent readout. We do not identify the two retained components with any physical doublet here; that reading belongs to downstream work that takes $\mathcal{R}_\chi$ as input.

Definition 10.4 (Artin phase and scale-coherence features). For $y = \mathcal{F}_\varphi f$, the Artin phase vector is $\Delta\varphi_j = \arg(y_{j+1}/y_j)$, $j = 0, \dots, 8$, with golden target $\tau_j = \varphi^{j/5}$. The FMT–DFT contrasts are $\Delta\text{TV} = \text{TV}_{\text{DFT}} - \text{TV}_{\text{FMT}}$ and $\Delta R^2 = R_{\text{FMT}}^2 - R_{\text{DFT}}^2$.

Remark 10.5 (Correct discriminant). $R_{\text{FMT}}^2 \approx 1$ for all test signals including harmonic ones, because the 4.70° Vandermonde arc produces smooth phase increments regardless of signal type. The operative discriminant is the contrast ΔTV and ΔR^2 , not R_{FMT}^2 alone.

11 Benchmarks

The $\Delta G < 0$ result is structurally expected: the 4.70° Vandermonde arc makes FMT basis functions nearly collinear, so no smooth signal can be sparse in the FMT basis. This does not diminish the two-sheet and doublet-structure results; it defines the domain of applicability precisely.

Table 6: Scale-coherence diagnostics. Primary columns: ΔTV and ΔR^2 . $R_{\text{FMT}}^2 \approx 1$ for all signals (Remark 10.5); use contrasts to discriminate.

Signal	T	R^2	TV	ΔTV	ΔR^2	$ \overline{S_z} $
Fib. anyon braid	FMT	0.997	0.070	11.95	0.536	0.0071
Fib. anyon braid	DFT	0.461	12.02			
Commuting trivial	FMT	1.000	0.031	18.82	0.677	0.0071
Commuting trivial	DFT	0.323	18.85			
Harmonic (control)	FMT	1.000	0.035	13.14	0.199	0.0071
Harmonic (control)	DFT	0.801	13.17			

Table 7: Gini coefficient (spectral sparsity). DFT wins uniformly. The FMT is a scale-coherence diagnostic, not a compression codec.

Signal	G_{FMT}	G_{DFT}	ΔG
Harmonic	0.280	0.800	−0.520
Log-periodic	0.156	0.607	−0.451
Golden pulses	0.027	0.582	−0.555
Fib. qubit readout	0.019	0.586	−0.567

12 Limitations and Future Work

The present paper fixes the golden ratio $\rho = \varphi$, the cell size $N = 10$, and the LOW/HIGH two-sheet readout. These choices define a reproducible scale-transfer frame, but they are not claimed to be optimal for all signals. A systematic ablation against dyadic band ratios, wavelet-packet ratios, standard log-Mellin grids, plain two-band splits, and learned projections is required before any application-specific optimality can be claimed.

The diagonal-unitarization result should be interpreted narrowly. Since $\mathcal{F}_\varphi$ is a square Vandermonde matrix with distinct nodes, an exact global non-diagonal metric exists,

$$W_{\text{glob}} = \mathcal{F}_\varphi^{-1}(\mathcal{F}_\varphi^{-1})^* = (\mathcal{F}_\varphi^* \mathcal{F}_\varphi)^{-1}, \quad \mathcal{F}_\varphi W_{\text{glob}} \mathcal{F}_\varphi^* = I \quad (27)$$

in exact arithmetic (the two expressions are the same matrix for an invertible $\mathcal{F}_\varphi$). This identity is not a practical normalization rule in double precision: the frame has $\kappa(\mathcal{F}_\varphi) \approx 4.7 \times 10^{18}$, so the normal matrix $\mathcal{F}_\varphi^* \mathcal{F}_\varphi$ has condition number of order 10^{37} , and its inverse cannot be formed reliably in standard precision (we confirmed the identity only at 80-digit precision). The global metric is dense, non-local, and numerically ill-conditioned. The nontrivial obstruction proved in Proposition 8.1 is therefore *local*:

no real diagonal reweighting of the lattice nodes turns the frame into a Parseval frame. The useful readouts are thus either the raw non-unitary scale-transfer ratios or the polar unitary factor $U_\varphi = WV^*$, defined canonically in exact arithmetic and evaluated numerically through the SVD. Since $\kappa(\mathcal{F}_\varphi) \approx 4.7 \times 10^{18}$, the orientation of the smallest-singular-value subspace is sensitive to finite-precision perturbations; unitarity of the computed factor does not by itself certify backward stability with respect to the exact frame. This is not a residual caveat but a measured effect: the double-precision factor differs from the exact one by $O(1)$ in Frobenius norm (§9.1). The specific U_φ and $P_\chi = \mathcal{R}_\chi^* \mathcal{R}_\chi$ are therefore computed in multiprecision, while the abstract coisometry theorem remains exact for any unitary U_φ .

A generalized ρ -Mellin transform, in which the scale ratio ρ is estimated from the signal rather than fixed to φ , is a natural extension for self-affine or material-specific scale structures. Dedicated fast algorithms for the golden finite frame are not proved here; the present implementation uses fixed-size cells and relies only on standard linear-algebra operations. External validation on independently labelled scale-structured datasets — for example, applying the scale-coherence features to Fibonacci-driven experimental data — also remains future work.

13 Conclusions

The Fibonacci–Mellin Transform has a natural LOW/HIGH two-sheet structure: the raw output is $y_j = \text{LOW}_j$ and $y_{j+5} = \text{HIGH}_j$, where the HIGH row is the LOW row pointwise Artin-twisted before projection ($(\mathcal{F}_\varphi)_{j+5,k} = A_j(k)(\mathcal{F}_\varphi)_{j,k}$) via the Fibonacci identity $1 + \varphi^{-1} = \varphi$. The D_5 -symmetric graph $\text{GP}(5,2)$ (the Petersen graph, vertex-transitive but not a Cayley graph) is the indexing model of this two-sheet structure: outer pentagon = LOW, inner pentagon = HIGH, radial edges = the entry-wise phase twist we label “Artin twist”. This name denotes a per-index phase factor only; it is *not* a braid-group/Artin representation and carries no topological-computing claim. The optional $\text{CT}(2,2)$ post-processing creates symmetric/antisymmetric channels $c_j^\pm = y_j \pm y_{j+5}$; it is not part of $\mathcal{F}_\varphi$.

Grouping the two sheets into the scale doublet $\Psi_j = (\text{LOW}_j, \text{HIGH}_j)^T$, the local phase cell $G_j(k) \in U(2)$ and $\tilde{G}_j(k) \in SU(2)$ after global phase removal. The FMT thus exposes a $U(1) \times SU(2)$ -type readout structure not present in the canonical scalar-bin DFT readout unless an extra pairing is imposed; it is a structure of the readout, not a physical symmetry. The Stokes-like observables (S_x, S_y, S_z) on the doublet measure cross-sheet coherence and energy imbalance; together with ψ_j and ΔR^2 they form candidate feature inputs for scale-drift analysis and classification support (validation against standard DSP estimators is required for application-specific claims).

The exact structural results are: the LOW/HIGH two-sheet decomposition with its Artin-twist relation (Theorem 3.1); frame non-tightness with a quantified Euclidean isometry deficit (Proposition 2.3); the polar unitary readout $U_\varphi = WV^* \in U(10)$

(Theorem 9.2); and the coherent rank-one reduction $\mathcal{R}_\chi : \mathbb{C}^{10} \rightarrow \mathbb{C}^2$ with $\mathcal{R}_\chi \mathcal{R}_\chi^* = I_2$ and $\dim \ker \mathcal{R}_\chi = 8$ (Theorem 10.2), so the coherent quotient is canonically two-dimensional. Numerically certified for $N = 10$ are: the obstruction to real diagonal unitarisation (Proposition 8.1); the condition number, effective numerical rank, and tested phase non-commensurability. Algorithm 1 delivers a scale-coherence feature vector; the primary discriminant is ΔR^2 , which separates φ -structured from harmonic signals across all tested classes.

The FMT is not a replacement for the DFT. It is a *scale-memory analysis transform*: it preserves the quasi-periodic scale advance $x_{k+10} = \varphi^2 x_k$, converts it into the two-sheet doublet structure, and reads out phase-coherence features that the DFT, operating on a closed additive cycle, cannot access. Separately from these signal-processing uses, the coherent rank-one reduction $\mathcal{R}_\chi : \mathbb{C}^{10} \rightarrow \mathbb{C}^2$ (Theorem 10.2) provides a self-contained operator-theoretic construction of a two-dimensional coherent quotient from the five-register two-sheet carrier; its origin in the rank-five SVD structure is established upstream [1], and its use as input to any further reading lies outside the present scope.

Declarations

Conflict of interest. None. **Funding.** None.

Code and data availability. All numbers in this paper are reproduced by two accompanying scripts: `certify_precision.py` (mpmath; the multiprecision certification of §9.1, single command, JSON output) and `qfmt_simulator.py` (NumPy/SciPy; frame, polar readout, benchmarks, figures). Both record the interpreter and library versions used in their output, and the distributed bundle carries SHA-256 checksums.

AI tools. Language editing only; author takes full responsibility for all mathematical content.

A Simulator

Certification (`certify_precision.py`, mpmath): singular spectrum, condition number, effective rank, polar factor U_φ and P_χ at 40–120 digits, and the double-precision comparison of §9.1.

Self-contained Python (`qfmt_simulator.py`): `build_fmt_matrix`; `polar_unitarization` ($U_\varphi = WV^*$ via SVD); `embed_unitary_to_power_of_two` ($U(16)$); `build_fmt_povm`; `halmos_block_encoding`; `compute_stokes_observables` (S_0, S_x, S_y, S_z per j); `artin_phase_metrics` (R^2 , TV, mono, ΔTV , ΔR^2); `scale_coherence_pipeline` (Algorithm 1).

B Operational Interpretation

The two transforms answer different questions

DFT: *which harmonic frequency is present?* ($k \mapsto k + N \equiv k$, closed additive cycle, scalar $U(1)$ bins).

FMT: *where is the signal inside a φ -autosimilar scale hierarchy, and how did it advance from one scale level to the next?* ($k \mapsto k + 10 \Rightarrow x_{k+10} = \varphi^2 x_k$, quasi-periodic multiplicative cycle, $U(2)$ doublets).

Three practical uses

1. Scale-drift detection. If a signal shifts multiplicatively along $x_k = \varphi^{k/5}$, the DFT spreads energy; the FMT Artin phase remains monotone. Domains: log-periodic precursors, quasi-periodic drives, Fibonacci photonic structures, discrete scale-invariant failure signals.

2. φ -rational phase feature extraction. The doublet observables (S_x, S_y, S_z) and the inter-sheet phase $\psi_j = \angle(\text{HIGH}_j/\text{LOW}_j)$ are features absent from the DFT. For damage detection, one may test whether healthy structures produce stable doublet trajectories and whether damage perturbs (S_x, S_y, S_z) ; this remains an open validation problem against MAC criterion, SSI, PCA, and Mahalanobis indicators (see Table 1).

3. Classification support. Harmonic: $\Delta R^2 \approx 0.2$, $\Delta \text{TV} \approx 13$. φ -rational: $\Delta R^2 > 0.5$ for all classes tested (log-periodic, scale drift, Fibonacci drive). ΔTV is class-dependent: large for Fibonacci/braid-like signals (> 8), smaller for log-periodic (≈ 2). ΔR^2 is the more robust discriminant across all φ -structured types.

The FMT is not a DFT replacement

DFT is optimal for: harmonic signals, spectral compression, OFDM, standard frequency estimation.

FMT is the natural tool for: multiplicative lattices, log-periodic signals, discrete scale invariance, scale-drift tracking, quasi-periodic drives. Candidate domains (validation open): OMA/SHM log-scale coherence, fractal/self-affine textures.

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Table 8: Scale-memory diagnostics (four signal types).

Signal	FMT TV	DFT TV	ΔTV	ΔR^2
Harmonic	0.035	13.17	13.14	0.199
Log-periodic	0.025	2.10	2.08	0.904
Scale drift	0.045	1.40	1.36	0.532
Fibonacci drive	0.020	8.97	8.95	0.929

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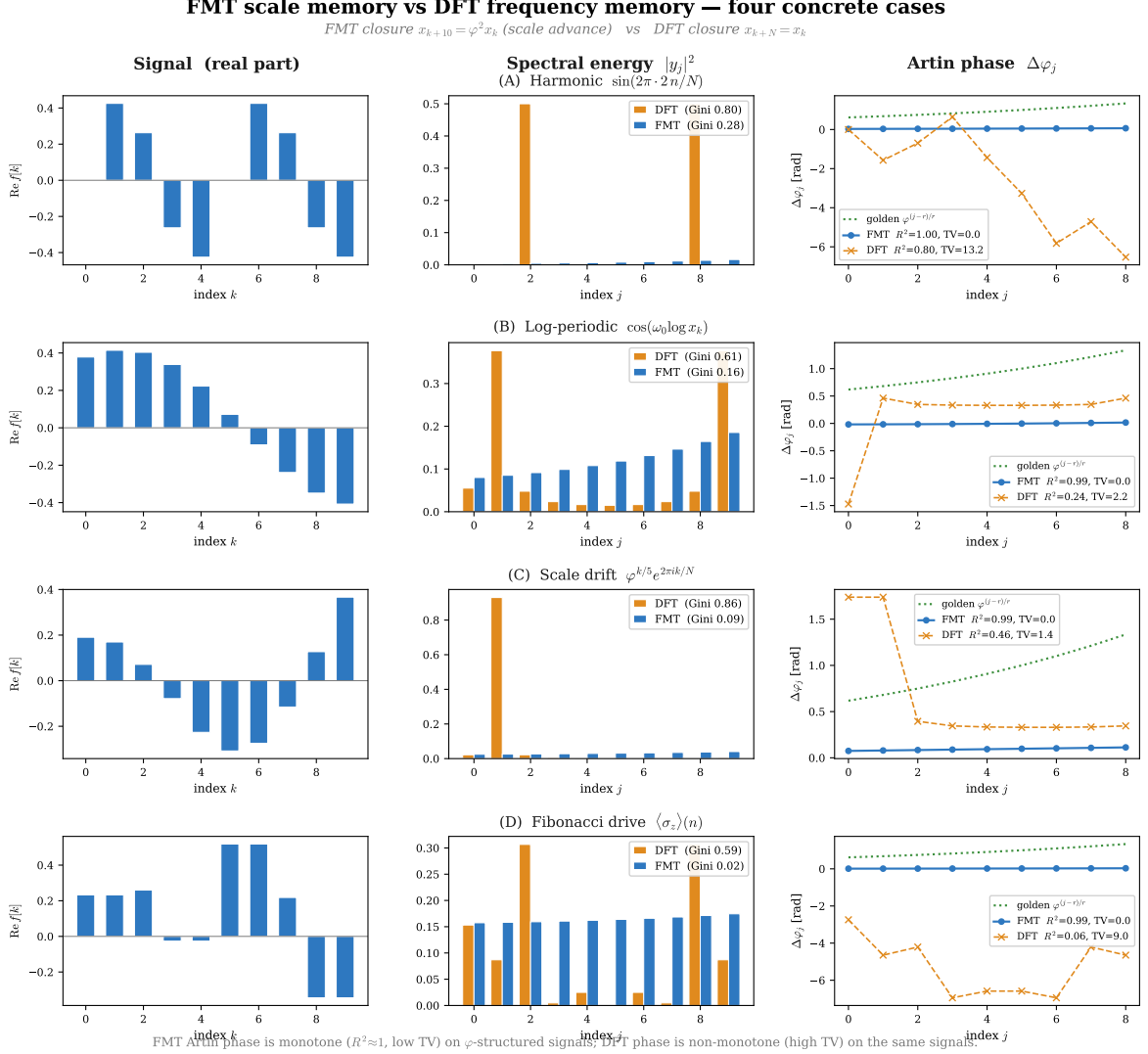


Figure 4: Scale memory demonstration (four signal types). Columns: signal on golden lattice $x_k = \varphi^{k/5}$; spectral energy FMT vs DFT (Gini); Artin phase $\Delta\varphi_j$ vs golden target $\tau_j = \varphi^{j/5}$. Primary diagnostic: ΔTV and ΔR^2 (FMT–DFT contrast), not R^2_{FMT} alone.